

Grade 7
Unit 10
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Josephine paid \$650 to rent a beachfront cabana at a vacation resort. For each guest that used the cabana a \$19.40 resort fee was added to the bill. If the bill was \$882.80, how many guests used the cabana?

1. What was the total amount of money spent on the cabana?

\$882.80

2. Which cost depends on the number of people who use the cabana?

\$19.40 resort fee per guest

3. Which cost is the same, regardless of how many visit the cabana?

\$650 cabana rental fee

The total cost for the cruise can be represented by the equation $882.80 = 19.40g + 650$.

4. What does the g represent in the equation?

The number of guests on the cruise.

5. What does the variable term, $19.40g$, represent in the equation?

The cost per guest.

6. What does the constant term, 650, represent in the equation?

The one-time cost for the cruise.

7. Solve the equation $882.80 = 19.40g + 650$

$$\begin{array}{r} -650 \qquad \qquad -650 \\ \hline 232.80 = 19.40g \\ \underline{19.40} \quad \underline{19.40} \\ g = 12 \end{array}$$

A local electrician charges a \$55 fee for a work order and \$78.04 per hour while they work. If the total bill was \$289.12, how many hours did the electrician work?

8. Write the equation that represents this situation using the variable h for the unknown quantity.

$$78.04h + 55 = 289.12$$

9. Solve the equation.

$$\begin{array}{r} 78.04h + 55 = 289.12 \\ -55 \qquad -55 \\ \hline 78.04h = 234.12 \\ \underline{78.04} \quad \underline{78.04} \\ h = 3 \end{array}$$

10. Explain what the solution means in this context.

The electrician worked for 3 hours.